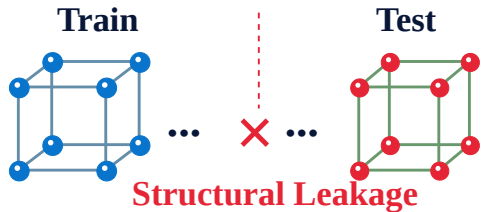


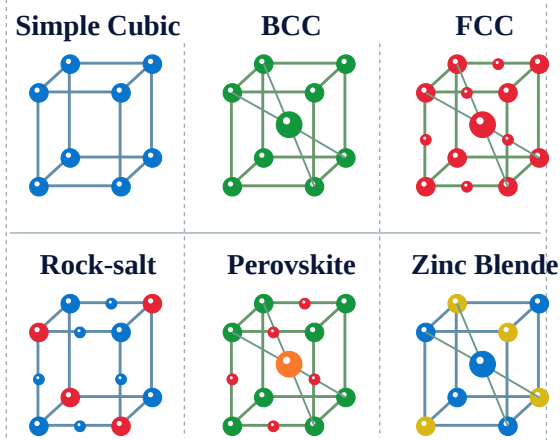
1. Crystal Dataset $\mathcal{D}$

Random Split (Preliminary)



StructureMatcher

Structural-Equivalence Graph $\mathcal{G}_{eq}$

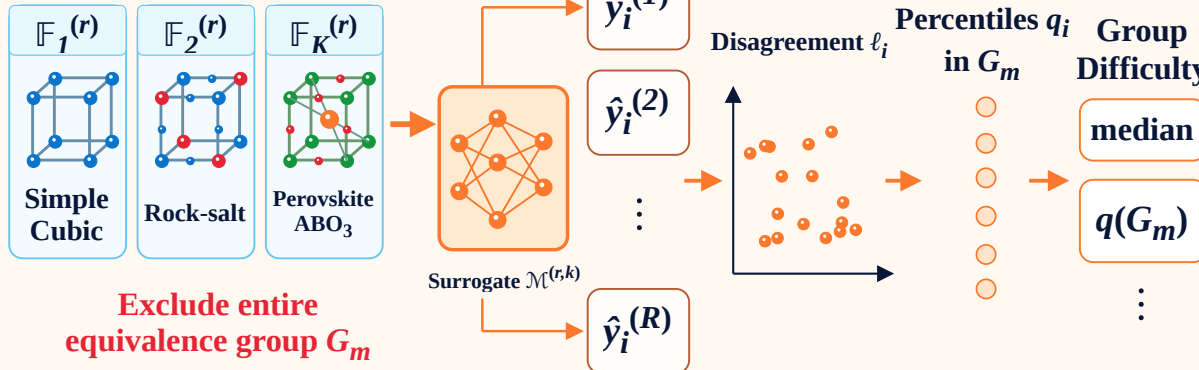


Connected Components $G_1 \dots G_M$

**No Equivalent Structures
Across Dev./OOD Boundary**

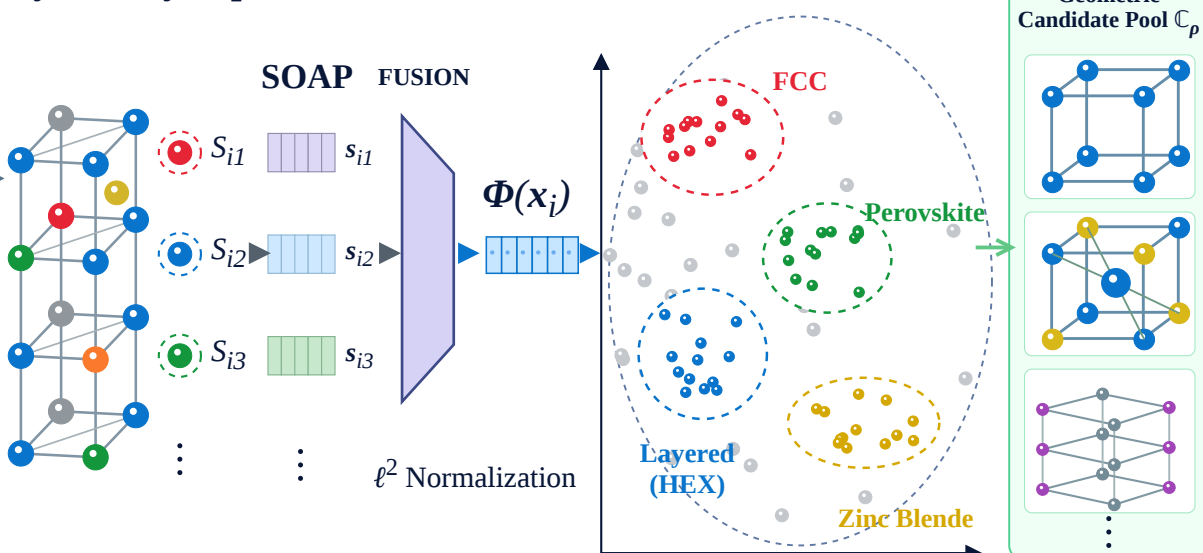
2. Group-Aware Cross-Fitting for Predictive Difficulty Estimation

Training on remaining folds
(exclude $\mathbb{F}_k^{(r)}$)



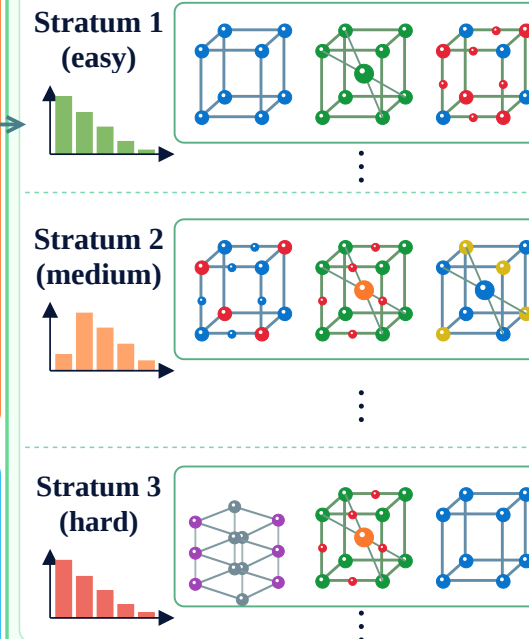
3. Structural Embedding and Global Geometric Isolation

Symmetry-Equivalent Sites

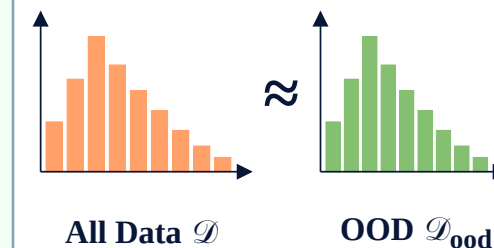


4. Difficulty-Calibrated Selection

Difficulty Strata $\mathbb{B}_1, \dots, \mathbb{B}_B$

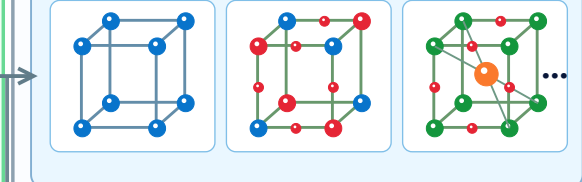


Difficulty Distribution Preserved

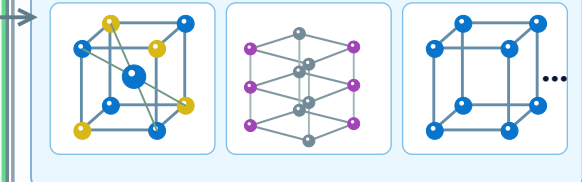


5. Leakage-Controlled OOD Benchmark

$\mathcal{D}_{train}$



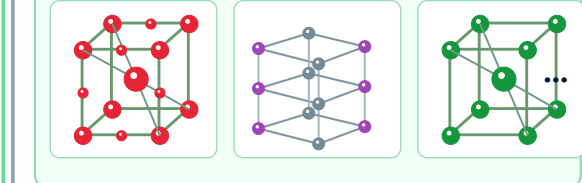
$\mathcal{D}_{val}$



No Equivalent-Group Leakage

Controlled Structural Shift

$\mathcal{D}_{ood}$



Split-Quality Diagnostics

